

Essentials to a good science paper

- Main sections:
 - Title
 - Authors and Affiliations
 - Abstract
 - Introduction
 - Observations
 - Analysis
 - Figures
 - Tables
 - Results
 - Discussion
 - Conclusions and Future Work

General guidelines

- Dry writing – not meant to be superfluous
- Succinct writing – avoid verbosity, be clear and concise. Clarity is important: consider reading each sentence out loud in a paragraph until it flows well when spoken. Shorthand and acronyms can be used, but must be defined the first time they appear.
- Do not make claims that are not justified → every claim either was done by you OR, has an origin in a paper you are citing / referencing, but don't take this to an extreme. What might be regarded as common knowledge to your audience – your peers or a professional scientist in our subfield does not need a citation.
- Logical writing. There should be a linear flow of the logic in the writing; things build upon others. Foundations get laid and built upon. Almost like a computer program.
- Avoid qualitative qualifiers
- Avoid sentences that do not advance the narrative – avoid sentences that are “meaningless mouthfuls of nothing”. Avoid phrases that start sentences that carry no meaning, such as “It is straightforward that...”
- Use the word “We”
- Try to write in the active voice and not a passive voice.
- Try to use consistent tense; observations and analysis can be past tense, or can be present tense.
- Include what I like to call “connective tissue” to help the paper flow between sections, such as the final paragraph of the introduction that introduces what each section will be about, or the first paragraph of each section which introduces what each sub-section is about.

What to write it in?

- Easy approach: Microsoft Word or Google Docs
 - I turn in grant proposals with word and google docs, and Nature and Science take Word paper submissions
- Harder approach: User Overleaf with a free account to prepare a paper in latex using the aastex6.3.1 template:
<https://journals.aas.org/aastex-package-for-manuscript-preparation/>
 - (download and put into the overleaf project directory)
 - Latex is like a half-century old version of html that is used for document formatting language. It continues to this day to be used for many physical science publications because of its superior equation, figure and table capabilities.



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Title

- Keep it Simple: example:
- “Ground-based light curve follow-up validation observations of TESS object of interest TOI NNNN.01”
- Authors, you, me. Use my affiliation:
 - Department of Physics and Astronomy, 4400 University Drive MS 3F3, George Mason University, Fairfax, VA 22030, USA





Abstract

- This works a bit like an outline of your paper, and is what most people will read because they won't have time to read the whole paper. It is best to be succinct and follow a fairly formulaic structure. In fact, European Astronomy Journals tell you to have the following structure in abstract with 2-3 sentences each for: Context (background), Aims (goal of investigation), Methods (how it was carried out), Results (what was found).



Introduction

- This is the hardest section for any new scientist to write when they are preparing a properly done professional science paper.
- Why is it hard? The introduction lays the foundation for your paper. It motivates why it should exist in the current context of the other literature that has been done in the recent past that is complementary and scientifically “near” the work you are doing, but is such that your work is still needed.
- A formulaic set of paragraphs can be as follows. This is not meant as a requirement, but a rough guide:
- Para 1 – the broad overview of the subfield – exoplanet transits for example – introduce TESS and candidate validation process
- Para 2 – dive deeper into the subfield to start motivating why your paper should exist – e.g. there are thousands of candidates that need to be validated. Similar papers, such as the examples cited in your “homework assignment” at the end of today, place your work in the context of existing but similar work. Here it is a good idea to provide quick 1-2 sentence summaries of a few papers most similar to yours for context.
- Para 3 – This paragraph now directly starts explaining the need of your paper. Oftentimes, early work leaves some questions unanswered or opens new questions. These are not shortcomings of the previous papers, so do not phrase them in a negative way. Besides, the authors of those papers won’t like having their paper talked about negatively. They worked with the state of the art and the state of the field at the time, and your paper is advancing that knowledge, addressing the questions they opened or left for a future work (see your own paper conclusions for example). The state of the field has now advanced to your paper. In particular for your paper, often there has not been previously published a paper on your particular TESS Object of Interest. So there is a need for more follow-up observations. Be careful here not to get ahead of yourself into the next paragraph.
- Para 4 – this is the paragraph that you want to start: “In this paper, we present follow-up observations of TOI XXXX.” Here you can say a little about your target – what is it’s radius, orbital period? What type of star does it orbit. But keep it light and not too detailed - that comes later. What are your goals? Our goal is to investigate whether or not the transit occurs on the expected star at the expected time, with the expected duration and depth.



Introduction

- Para 5: This is a very dry paragraph that I like to call “connective tissue”. It introduces the sections of the paper, and is very literal, almost like a table of contents, but in written form:
 - “In Section 2, we present our Observations from TESS and the George Mason University 0.8m telescope. In Section 3, we present our analysis of the TESS light curve for TOI XXX.01 and our ground-based light curve analysis. In Section 4, we present our light curve results. In Section 5 we discuss our results and in Section 6 we present our conclusions and future work.”





Observations Section

- The section starts with a pre-amble, a paragraph that is again a very dry introduction to this section of the paper that is “connective tissue” that helps join the flow of the paper sections in a logical flow:
 - ”In Section 2.1 we present the TESS Object of Interest XXXX.01 and its exoplanet candidate properties, its host star properties from the TESS Input Catalog, the *Gaia* mission, and other archival sources. In Section 2.2 we present the TESS sector light curve(s). In Section 2.3, we present a summary of the observational data collected with the George Mason University 0.8m telescope.”



Observations Section

- Then this section has a set of sub-sections that describe the observational data collected. It is important here to not talk about any analysis or results. It is a data accounting section. The rest comes later. If there is some data processing that is done prior to your analysis – for example, the aperture photometry performed by the TESS mission itself to generate your light curve, that is ok to talk about here, since it occurred by another team prior to your work. It is important to add a citation here.



Observations Section

- How many exposures?
- What was the exposure time?
- What was start/end times?
- [if not you, then add the students that collected the data as co-authors, so you can still say “we”]
- What were the dates?
- What filter did you observe in?
- What was the RA and Dec of your target?
- This very much a log of information about the data collected.



Analysis Section

- The section again starts with a pre-amble, a paragraph that is again a very dry introduction to this section of the paper that is “connective tissue” that helps join the flow of the paper sections in a logical flow:
 - ”In Section 3.1 we present our tools used to analyze the TESS sector light curve(s) curves using AstroImageJ/ExoFASTv1/ExoFASTv2. In Section 3.2, we present our analysis of the ground-based light curve using AstroImageJ.”
- Again, each subsection goes into detail of the analysis done. The goal here is reproducibility that is a hallmark of the scientific method. You want the reader to be able to reconstruct and reproduce the analysis that you did on their own.
- There is a trade between providing too much detail vs too little detail. You can assume your reader is an expert. If there is a document you can refer to as a citation for a more detailed explanation of steps, you can cite it there (for example the paper on ExoFAST if you are using that).
- Again, try to limit your presentation to just analysis here; don’t mention results or discuss results, that comes later.
- Here some figures and tables are useful, and see later slides on that.



Results Section

- Once again, this starts with a dry connective tissue paragraph here like the previous sections.
- Be very careful here to only present results. Do not discuss or interpret them. That goes into the next section. Do not present additional analyses; that belongs in the prior section.
- Oftentimes just some tables and figures and explanations thereof in the text portion will suffice for a result section. It can be very simple in that regard.





Discussion Section

- Ah finally, now we get to discuss what we've learned in our project.
- Here again, we start with a connective tissue paragraph:
- “In Section 5.1 we present our interpretation of our results. In Section 5.2 we place our results into context of the greater field of follow-up of candidate exoplanets from the NASA TESS mission.” (note you can tailor this based upon what you've found! You don't know your results yet!)

Discussion Section

- In 5.1 – you might want talk about whether or not you have a detection of a false-positive? If a detection, were timing uncertainties large such that you provide a better knowledge of the timing (ephemerides). If a non-detection – why? Was it cloudy? Was the data too noisy so that the transit depth was detectable? Was an eclipsing binary found nearby? Did the transit depth match what was expected from TESS? If not, why not?
- In 5.2 – you want to compare your results to the work of others? Is this a hot Jupiter? What kind of planet is it? Do we know of other planets like this? Is there another example, published or not, of observing this hot Jupiter, or not?
- This can be a pretty hard section to write, because you can't start it until you know your results, and what you were expecting.
- Kevin and I will provide for you what the TESS team knows about the candidate your observed, and you compare what you got to that knowledge.
- For your first scientific paper, it's ok if this not fully-fleshed out. This is also hard for a new science writer to right, specifically section 5.2, because you don't know much about the field.



Discussion Section

- As a bonus section – one can go through more complicated analysis to statistically rule out false-positives using codes like vespa or similar, but that is beyond the scope of this internship. But if you want to try, you're welcome to do so.
- So, the ultimate goal of ground-based follow-up is to enable the scenario where there is a $<1\%$ chance that your candidate is a false-positive. And that is hard to do from the light curve alone. One has to use the NASA Exoplanet Archive Exoplanet Follow-Up program website – <https://exofop.ipac.caltech.edu/> - to gather and present additional data that we discussed last week – high contrast imaging to see if there are any faint nearby stars, spectroscopy to see if there are any binaries present in the spectra and rule out large radial velocity changes; and light curve tests on the TESS light itself that come from the TESS mission.

Conclusions and Future Work Section

- This is similar to the abstract, but is not written in hindsight and focused more on the results and discussion in the first paragraph.
- The second paragraph is usually the future work that discusses the loose ends – for example, many of you will not carry out the detailed statistical false-positive validation analysis, so that be left as future work, and the light curve will be one piece of helping to validate the existence of the candidate exoplanet.
- In some cases you might have definitive conclusion if it is a false-positive – then there is no future work – it is an eclipsing binary for example and someone else might be interested in studying for studying binary star properties for example.

Figures and Tables

- Figures should be legible – take figures from AIJ or python’s matplotlib or ExoFAST (convert postscript “.ps” files to PDF files in that case).
- Figures should be lossless file types – NO JPEGs. PNGs preferred. And vector image graphics (can be scaled to any number of pixels) are preferred over rasterized images (pixelated at a fixed pixel scale): See for an explanation:
<https://pavilion.dinfos.edu/Article/Article/2223089/vector-vs-raster-images-choosing-the-right-format/>
- Figures have captions. In Nature/Science, figure captions carry some of the story telling for further brevity. In other journals, the captions are dry and explanatory (The horizontal axis is blah, the vertical is blah, the data points shown in green are blah, etc.), but refer to the main text to tell the story (“See section blah for a discussion of this figure”).
- Figures and tables are numbered. In latex with aastex631, there are special ways to make figures and tables, ask and I can send you examples if you go this route.
- There are examples of figures and tables provided in your ”homework assignment”
- These can be placed throughout paper –usually starting in the observations section and going through results. There is rarely if never any figures/tables in the Intro, and only rarely in the discussion.

References

- References are included throughout the paper as citations. As an example, you need a citation when you make a claim that is not justified in your work directly, or refer to a process that was developed elsewhere.
- For example: “There are over 5000 exoplanets discovered or validated to orbit other stars to date (Akeson et al. 2013).” (great opening for a first sentence of an intro)
- You can also cite a website directly: “There are over 5000 exoplanets discovered or validated to orbit other stars to date (<https://exoplanetarchive.ipac.caltech.edu/>).” or you can do it as a footnote if it’s a website:
- You can also cite a website directly: “There are over 5000 exoplanets discovered or validated to orbit other stars to date¹.” at the bottom of the paper page:
- 1 <https://exoplanetarchive.ipac.caltech.edu/>

References

- Citations can either be in parentheses or not (in line with the text):
(Akeson et al. 2013) , or as shown here: as shown in Akeson et al.
(2013)
 - Akeson is the first author's last name.
 - Et al. means all the other authors if more than three
 - If 2: "Plavchan & Berberian 2023"
 - If 3: "Plavchan, Currie and Simpson 2015"
 - 2013 – is the year of publication
- Finally, at the end of the paper, all citations that were in a paper are listed again with more detail. See the example papers provided.
 - Here you alphabetize (or you put in order of citation):
 - Akeson, R.L., et al., "title of the paper", Proceedings of the Astronomical Society of the Pacific, 2013, volume NN, page NN
- Ask for help on how to do these, the formatting is not rigorous for this internship!